THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 1: **Patterns in Mathematics** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. What is the sum of the first 20 natural numbers $1 + 2 + 3 + \cdots + 20$? 1

(a) 210 **(b)** 190 **(c)** 230 **(d)** 211

2. The numbers 1, 4, 9, 16, 25, ... are called square numbers. The 17th one is: 1

(a) 306 **(b)** 289 **(c)** 272 **(d)** 290

3. In the pattern $1, 3, 6, 10, \ldots$ what is the 12th term? 1

(a) 78 **(b)** 90 **(c)** 79 **(d)** 77

4. In the pattern of cube numbers $1, 8, 27, 64, \ldots$ the term after 216 is: 1

(a) 353 **(b)** 342 **(c)** 343 **(d)** 344

5. In the Virahanka (Fibonacci) sequence $1, 2, 3, 5, 8, \ldots$ the 9th term is: 1

(a) 56 **(b)** 54 **(c)** 57 **(d)** 55

6. A complete graph has 5 points, each joined to every other point. How many lines are there? 1

(a) 9 **(b)** 11 **(c)** 10 **(d)** 15

7. The sum of the triangular numbers 21 and 28 is: 1

(a) 48 **(b)** 50 **(c)** 49 **(d)** 55

8. What is the sum of the first 25 natural numbers $1 + 2 + 3 + \cdots + 25$? 1

(a) 325 **(b)** 326 **(c)** 350 **(d)** 300

9. The numbers 1, 4, 9, 16, 25, ... are called square numbers. The 19th one is: 1

(a) 342 **(b)** 380 **(c)** 362 **(d)** 361

10. In the pattern $1, 3, 6, 10, \ldots$ what is the 13th term? 1

(a) 104 **(b)** 91 **(c)** 92 **(d)** 90

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. How many lines are needed to join all pairs among 9 points (a complete graph)? 2

12. Find the sum $1 + 3 + 5 + \ldots$ of the first 15 odd numbers without adding one by one. 2

13. Write the first 10 terms of the Virahanka sequence that starts with 2, 3. 2

14. How many lines are needed to join all pairs among 10 points (a complete graph)? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Add the triangular numbers 66 and 78. What do you notice about the answer? 3

16. Write the sum $1^3 + 2^3 + \cdots + 5^3$ and show it equals the square of $1 + 2 + \cdots + 5$. 3

17. Look at the sequence $11, 15, 19, 23, 27$. Find the pattern and the 10th term. 3

OR

Add the triangular numbers 78 and 91. What do you notice about the answer? 3

18. Write the sum $1^3 + 2^3 + \cdots + 6^3$ and show it equals the square of $1 + 2 + \cdots + 6$. 3

OR

Look at the sequence $1, 10, 19, 28, 37$. Find the pattern and the 10th term. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A group of 8 friends shake hands with each other exactly once. Using the idea of a complete graph, find the number of handshakes. Show the working. 5

OR

In the Virahanka sequence $1, 2, 3, 5, 8, \ldots$ the 7th and 8th terms are 21 and 34. Find the next three terms and add the first 7 terms. 5

20. A group of 9 friends shake hands with each other exactly once. Using the idea of a complete graph, find the number of handshakes. Show the working. 5

OR

In the Virahanka sequence $1, 2, 3, 5, 8, \ldots$ the 8th and 9th terms are 34 and 55. Find the next three terms and add the first 8 terms. 5

END OF QUESTION PAPER — Total: 40 Marks

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